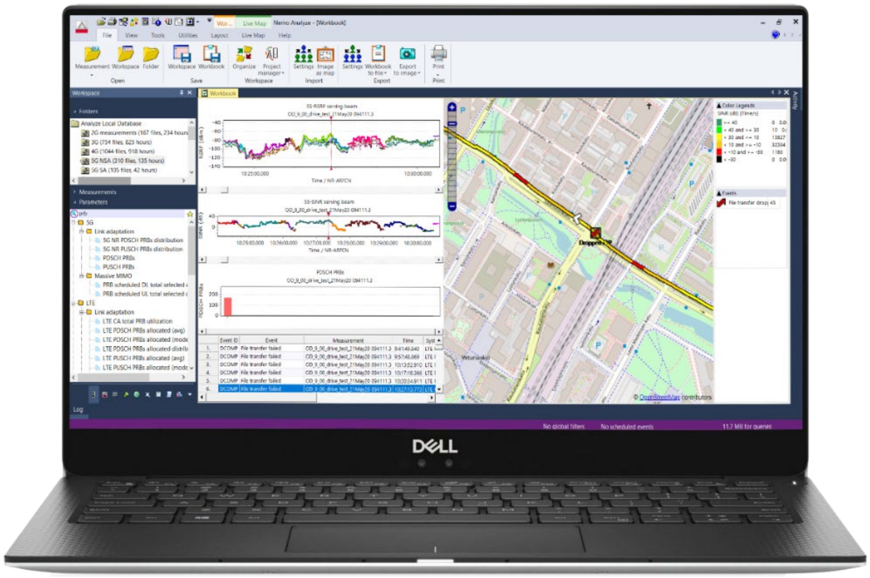
Nemo Analyze

Professional and Powerful Post-Processing of Field Test Data

Nemo Analyze is a cutting-edge analysis tool for performing optimization, troubleshooting, benchmark analysis, and statistical reporting. Nemo Analyze helps you to streamline your processes by enabling the automation of the entire data processing chain from a field test to an open workbook with analysis results. The system incorporates an innovative, low-maintenance database engine that has been designed and optimized specifically for high-performance post-processing of field test data.

Nemo Analyze offers a comprehensive set of technology-specific Key Performance Indicators (KPIs) for the latest wireless technologies and a wide range of data views that are known to offer the best visualization of field test data on the market — and yet it is highly cost-effective, easy to set up and use, and it scales to meet the needs of organizations of any size.

Nemo Analyze supports QoS/QoE analysis of all applications supported in Nemo measurement tools, e.g., voice calls with voice quality, YouTube, FTP, and HTTP. Nemo Analyze offers support for all major wireless technologies, including 5G NR, LTE-A CA, NB-IoT, LTE-M, VoLTE/ViLTE, VoWiFi, and mMIMO. NPS (Network Performance Score) with report and dashboard is also supported.



KEYSIGHT

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In addition to the Nemo measurement tools, Nemo Analyze supports the file formats InfoVista TEMS, PCTEL SeeHawk, and Rohde & Schwarz MNT (SwissQual) measurement tools. For more details, please contact our sales personnel at nemo.sales@keysight.com

Figure 1. Use Nemo Analyze to perform in-depth QoS/QoE analysis of measurement data from Nemo measurement tools.

Nemo Analyze – Fast and Efficient Post-Processing and Analysis of Field Test Data

Nemo Analyze Standalone is a desktop/laptop-based tool with a local, per-user database engine that runs on Windows 11 (64-bit). Nemo Analyze Enterprise is a client-server solution with a centralized multi-user database engine where the server side runs on Windows Server 2019/2022/2025. Nemo Analyze provides in-depth analysis, reporting, and optimization capabilities.

Nemo Analyze database engine combines the benefits of a heavy-duty database and a file-based analysis solution. Because of this, the performance of queries that produce vast numbers of rows matches that of file-based solutions and is outstanding compared to typical database solutions. From the maintenance perspective, the Nemo Analyze database functions exactly like file-based software and requires no administration. The only practical limit for the database size in the Nemo Analyze concept is the physical size of the hard disk.

With Nemo Analyze you can perform operations such as troubleshooting, map plotting, and report and workbook generation based on gigabytes of measurement data fast and efficiently. In addition, you can easily automate these routine operations; measurement data upload into the database and its generation into a report or a workbook can be scheduled to occur at any desired time on recurring intervals.

Nemo Analyze with its full set of graphs, maps, and other data views for detailed analysis and playback is the perfect solution for post-processing and statistical reporting tasks for users with specific KPI reporting and analysis needs. Each Nemo Analyze workbook can contain several pages and frames, such as a map, various graphs, layer 3 messages, decoded layer messages, and idle and/or connected mode KPIs. Workbooks, pages within the workbooks, and data views on each workbook page are all synchronized. Multiple layers in the same graph window offer a convenient way to compare two or more RF parameters simultaneously. Measurement data can be plotted on live maps (Google Maps and OpenStreetMaps) or MapInfo-compatible maps. With Nemo Analyze, you can perform root cause analysis and quick drilldown to problem details.

Accurate, Efficient Post-Processing of Field Test Data

The Nemo Analyze system incorporates a PostgreSQL database engine that has a strong reputation for reliability, data integrity, feature robustness, and performance. Easy to set up and use, this highly cost-effective tool provides you with predetermined report templates which allow you to compare KPIs from different operators, technologies, and time frames, and see the results in a single report.

Nemo Analyze also enables you to select the optimal approach for each statistical reporting task. A set of predefined MS Excel-compatible report templates for optimization and benchmarking purposes are available. The reports can be customized using the Nemo Analyze Spreadsheet Report Designer, and new custom reports can be easily created. Report templates can also be created for MS Word and MS PowerPoint with a limited set of features. Nemo Analyze also offers an extensive set of customizable, predefined workbooks and reports that enable effortless and thorough reporting based on all essential network and application-level KPIs, as defined by standards specifications such as ETSI. Reports can be run over multiple files and source data selected based on various attributes, including time frame (e.g., last week), system, cell, measurement file(s), Mobile Network Code (MNC), area, or any combination of the aforementioned.

Nemo Analyze enables the creation of custom KPIs and custom analysis tasks with KPI Workbench, a graphical flowchart-based scripting engine. Nemo Analyze includes automated analysis tasks for most common network problems, such as call drop root cause analysis, BCCH clash analysis, and missing neighbor analysis. Drilldown from problem events on maps and grids enables efficient troubleshooting over large sets of data and statistics and benchmarking based on operator, log file, time frame, and other attributes. Nemo Analyze enables troubleshooting the QoE of mobile networks by using mScore test results correlated with in-depth radio metrics available for post-processing.

5G NR Support

Figure 2. 5G beams visualized on a 3D map.

Nemo Analyze supports the analysis of 5G NR UE and scanning receiver measurements performed with Nemo data collection tools. Available metrics include dozens of 5G parameters and events from Nemo File Format. Nemo Analyze provides a comprehensive set of ready-made report templates and playback workbooks with all the key metrics and KPIs for quick analysis and an automated routine for plotting the SSB beam footprints of all beams is also included. With the optional 3D Visualizer, 5G NR beams can be visualized on a 3D model to detect the attenuation of buildings and trees on the signal level and to evaluate beam width and coverage in real life.

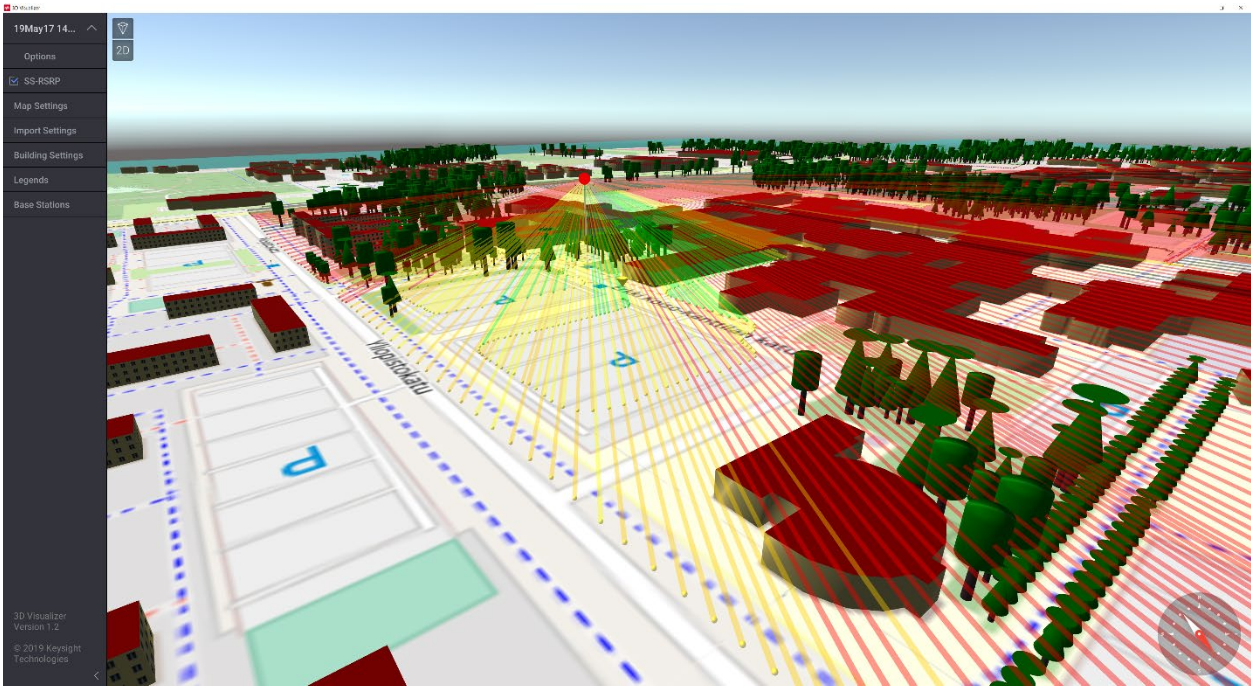


Figure 3. 5G beam visualization on OSM 3D model.

In addition, Nemo Analyze supports measurements done with Keysight´s FieldFox portable spectrum analyzer. The system provides the total channel power level over the measured bandwidth. Coupled with a continuous wave transmitter, the solution can be used for path loss measurements on the 5G NR frequencies from sub 6GHz to mmWave.

Nemo Analyze System Requirements

- Processor: Intel i7, 8 core or similar
- RAM: 32 GB RAM minimum
- Hard drive: 500 GB SSD (or high-performance HDD) minimum, 1 TB SSD recommended
- Ports: One USB port for HASP USB license key (optional depending on the license type)
- Operating system: Windows 11, 64-bit
- Installation requires administrator privileges

Nemo Analyze Enterprise Server System Requirements

System requirements are subject to change as more optimal server platforms become available.

- Operating system: 64-bit Windows Server 2019/2022/2025 (2025 recommended)
- Any Windows Server 2019/2022/2025-compatible server hardware
- Processor: Multi-core x64 compatible processors. We recommend that you use at least 8 cores. The overall data ingress and number of users may bring in additional requirements as for the number of cores.
- Memory: 32 GB minimum, 64 GB recommended. NOTE: If you run other services on the same computer, the memory requirement may be higher. For example, if Qlik Sense dashboards is installed on the same server, 128 GB is recommended.
- Storage:
  - Type: SSD, fault-tolerant disc is highly recommended (RAID)
  - System disk space: 100 GB minimum
  - Data disk space: The data disk space requirement depends entirely on the number of measurement hours the database should be able to contain to suit the needs of your organization. An estimate can be made based on the fact that a single measurement device produces approximately 15 MB of data per hour. In the server database, the data consumes ~55 MB per measurement hour. When the measurement files are backed up in compressed format and the data loaded into the database, 60-65 MB of storage space is needed per measurement hour.
- Connectivity:
  - The server needs to have a static IP address.
  - By default, the server communicates with clients via ports 12001-12003 and the database connection uses port 5435. Therefore, these ports must be open in the firewall(s). However, the server can also be configured to use other ports.
  - 100 Mbps bandwidth is the recommended minimum between clients and the server. Client-server communication works almost with any network speed, but a slow connection directly affects the performance and usability of the system from the end-user perspective. 1 Gbps Ethernet connection speed is recommended
  - The server must be able to make outbound connections on HTTPS port 443 to use Live Maps
- The number of potential simultaneous users should be considered when dimensioning the system. The server should have approximately 1 CPU core for every three users. With ten Nemo Analyze licenses, the adequate number of processor cores is four. There should also be 2 GB of additional memory per potential simultaneous user. With ten simultaneous users, the requirement is 20 GB of additional memory.

Licensing

Nemo Analyze Standalone and Enterprise Client support GLS (floating) and USB portable licensing.

Enterprise Server supports node-locked and USB portable licensing.

Nemo Analyze Key Benefits

- Newest technologies – Every major new wireless technology is always supported.
- Automated data processing chain – Streamline your processes with Nemo Analyze’s capability to automate the entire data processing chain from the upload of a field test campaign to a full report with detailed analysis results.
- Anomaly detection with artificial intelligence and machine learning.
- Seamless integration with Nemo measurement tools – Ensure 100% correct interpretation of the measurement data produced by your Nemo tools. Furthermore, Nemo Analyze is always the only tool in the market with full, completely up-to-date support for the Nemo File Format.
- Speed and scalability – Analyze and manage hundreds of hours of measurement data with best-in-class performance. Solutions by other vendors require expensive back-end server farms to match the scalability and performance of Nemo Analyze running on a single laptop.
- Diverse reporting options – Create and modify report templates with Microsoft Excel-compatible reporting.
- Best-in-class data visualization – Correlate enormous sets of parameters, events, and signaling data within a single workbook with Nemo Analyze’s fully customizable multipage, time-synchronized workbooks, and best-in-class graph options.
- Extensive map support – See your measurement routes with web-based live maps (Google Maps and OpenStreetMap).
- Proven, first-class technical support – Experience comprehensive and cost-efficient technical support from people who each have a thorough understanding of the entire Nemo product line.

Nemo Analyze Key Features and Functionalities

Enhance the reliability and accuracy of your wireless network with best-in-class data analysis and visualization.

Supports All Major Network Technologies

- Supported networks - 5G NR, LTE/LTE-A CA, NB-IoT, VoLTE, ViLTE, and VoWiFi, mMIMO, GSM, GPRS, EDGE, WCDMA, HSPA+, CDMAOne, CDMA2000, TETRA, TD-SCDMA, WiMAX™.
- Measurement tools - Entire Nemo measurement tool portfolio, InfoVista TEMS, PCTEL SeeHawk, and Rohde & Schwarz MNT (SwissQual).

Ensures High-Performance Post-Processing of Drive Test Data

- File-based, low-maintenance PostgreSQL database engine with outstanding performance compared to typical database solutions.
- In-building measurement analysis enables analyzing and post-processing in-building (i.e. indoor) measurements that have been performed on various floors of one or several buildings. Floorplans are supported in MapInfo and iBwave formats.
- Benchmarking of radio and application-level metrics based on various attributes over hundreds of hours of data.
- Correlation of uplink and downlink voice quality measurement files fully supported.
- Time-synchronized analysis of data connection measurements and IP/UDP packet trace data
- Support for web-based live maps (Google Maps and OpenStreetMap) map sources enable you to visualize buildings affecting the radio signal propagation in 3D, and to better estimate low coverage and interference areas.
- A comprehensive set of troubleshooting KPIs for 5G NR, LTE, UMTS, and GSM featuring, for instance, root cause analysis or cell footprint analysis.
- Automated problem survey with drilldown allows analysis from a general troubleshooting query to full event details.
- Automated detection/analysis of common GSM, UMTS, LTE, and 5G NR problems.
- Trend analysis
- Customizable base station reference files can contain any user-defined information element not defined in the Nemo BTS File Format, such as RNC ID, SGSN ID, and neighbor cells.
- Log file manager enables original log files to be searched and retrieved from the database (Enterprise Edition only).
- Queries can be limited to predefined geographical areas of interest using, e.g. imported MapInfo polygons.
- Reduces hard drive usage by up to 90% by automatically packing measurement files. A free hard disk space of 100 Gigabytes can contain up to 1 TB of measurement data.

- Data organization features make analysis tasks efficient. Organizing measurement data into folders or subsets and joining separate measurement files into a joined measurement makes it easy to focus on the analysis.

Provides Best-in-Class Visualization

- Synchronized workbooks, workbook pages, and data views
- A comprehensive range of data views, including maps, grids, line graphs, bar graphs, pie charts, surface grids, color grids, and spreadsheets
- Cutting-edge field test data visualization: area binning, distance binning, advanced base station overlay with the service areas, signal power, etc. of selected antennas visualized as area bins, and line drawing from terminal to active and monitored cells based on any parameter (e.g. indicating pilot pollution). Base station map overlay can be synchronized with base station data to enable automatic zoom to the map representation of the cell selected on the grid.
- Easy-to-use graphical interface for designing and creating custom KPIs without any knowledge of SQL.
- With 3D Visualizer option, 5G NR beams can be visualized on a 3D map to detect the attenuation of buildings and trees on the signal level and to evaluate beam width and coverage in real life.
- Playback of individual log files
- Dashboards (optional)

Automates Testing and Reporting and Provides Flexible Customization

- Full integration with Nemo Cloud enables you to automate the end-to-end process for analysis and reporting.
- Automatable data loading and report/workbook execution
- Automates scheduled workbooks, reports, and measurement data uploads from predefined folders. A notification email can be configured to be sent when a new workbook/report is generated, and when a defined KPI value changes.
- Pre-defined and customizable benchmarking reports, including CDR E2E voice and data reports where data is aggregated per transaction (regarding voice calls, both A and B parties are combined) and further analyzed to get the results.
- Statistical reporting with Spreadsheet Reports compatible with Microsoft Excel.
- Report templates can be created for MS Word and MS PowerPoint with a limited set of features.
- Flexible customization of user interface and many functionalities (workbooks, data view layouts, queries, color sets, KPIs, etc.). Custom settings can be exported and imported.
- Every operation performed with Nemo Analyze can be configured to be automatically filtered based on a set of parameter threshold conditions, enabling for instance data from outside coverage area to be filtered out of processed data sets.
- Custom SQL queries

Other Features

- Data export to MapInfo, Microsoft Excel, txt, and Google Earth
- Database filtering based on technology, time, operator, etc.
- Area binning
- Advanced cell reference information including drift from antenna main lobe
- Advanced data filtering and global filters
- ODBC connectivity to third-party databases
- ASCII import to database enables creating custom templates to import any flat character separated value data into the database.
- CSV ASCII import to database enables data to be post-processed and visualized using the data views available in Nemo Analyze, which facilitates the retrieval of external data, such as from Hawkeye.
- Keyhole Markup Language (KML) and KML Zipped (KMZ) supports color legends and altitude information, enabling users to view routes in 3D on Google Maps.
- Lee's criteria sampling for scanner measurements enables distance-based aggregation for scanner data.

Optional Features

- NPS – An optional feature consisting of a report for NPS (Network Performance Score). Based on established methods and metrics, the data collected in the benchmarking survey is weighted and aggregated per application, which finally results to a calculation of an overall score, allowing the ranking of the tested mobile networks.
- 5G Advanced Analytics – An optional feature consisting of analysis parameters and KPIs, including pilot pollution analysis, overspilling analysis, weak coverage analysis, bad quality analysis, and NSA 5G NR neighbor list analysis, to troubleshoot specific network issues in 5G networks.
- 3D Visualizer – An optional feature providing 3D environment model and visualization of field test data and beams.
- Binary log decoding for Qualcomm and Samsung diagnostic data.
- Advanced report automation. The optional feature for Nemo Analyze Enterprise enables fully automated reporting workflow.

With Nemo Analyze…

- You do not need to know any SQL or anything about the database solution. The entire functionality of the database is accessible to the user through Nemo Analyze's intuitive and easy-to-use graphical user interface.
- The required amount of database administration is little to none. Database installation and maintenance are easy. The innovative structure of the database and its built-in optimization for field test data makes tasks such as indexing and de-fragmenting unnecessary.
- The entire measurement file is stored in the database instead of just aggregates. It is never necessary to revert to the original measurement files.
- Measurement files are automatically packed to take only 10% of the space they would otherwise require. A free hard disk space of 100 Gigabytes can contain up to 1 TB of measurement data.
- The speed and performance of your database are outstanding compared to generic database solutions. The performance of queries over measurement files is good regardless of the database size because the file structure is maintained in the database.

- Your database size is limited only by the amount of free space you have on your hard disk. Hardware upgrades enable unlimited database size.
- You still get the benefits of using an SQL database. Large sets of data can be easily stored, managed, and processed. The open ODBC interface of the database ensures that the stored data remains accessible also to any third-party SQL database tool. Custom KPIs can be created also by using SQL.

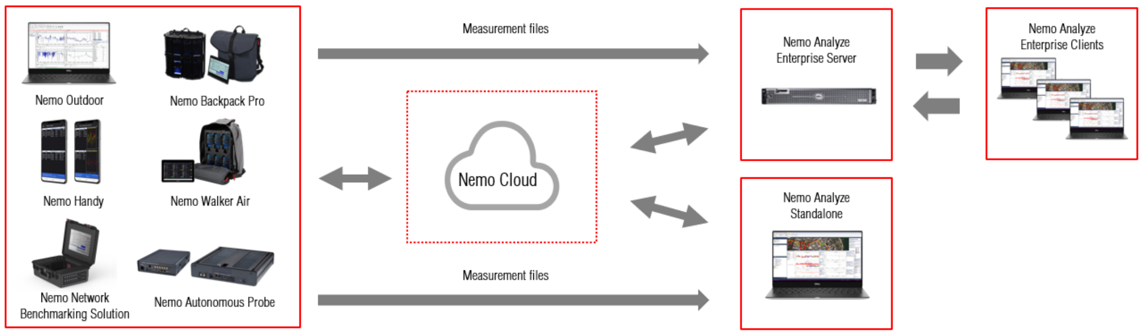


Figure 4. Nemo Analyze Database Concept

Anomaly Detection with Artificial Intelligence and Machine Learning

Anomaly detection helps in locating anomalies for efficient retesting, detecting rare anomalies, understanding operational problems and root causes. For each anomaly detection AI/ML will automatically identify and propose root cause. AI/ML can predict from the map exact GPS site where anomaly has happened. AI detects anomalies ML automates event classification. AI predicts beams, handovers, and mobile strategies. Supports both real-time and post-processing workflows and is available as a licensed feature. Below are the highlights of the feature

- Supports both real-time and post-processing workflows and is available as a licensed feature.
- Automatic anomaly detection
- Root cause analysis
- Reduced manual investigation
- Supports RAN and NTN measurements
- Improves troubleshooting speed
- Reduces drive testing costs
- Explainable AI results

Today’s mobile and satellite networks can be complex and data-heavy, creating a growing need for clear, actionable insights. Anomaly detection offers operators and vendors with below advantages,

- Detect rare and hidden anomalies in massive datasets
- Understand root causes, not just symptoms
- Reduce manual analysis and costly retesting
- Get role-specific insights across planning, deployment, and operations
- Proactively manage 5G and NTN network performance

Database Solution

Designing and optimizing the database specifically for field test data has enabled us to create a solution that not only manages to combine the benefits of SQL databases and file-based solutions but also avoids the common database bottlenecks that are all too often accepted as necessary trade-offs.

Data Views and User Interface

Nemo Analyze's highly intuitive user interface and multiple data views, together with its innovative concept of time-synchronized and customizable multipage workbooks provide you with the best visualization of field test data on the market. The range of data views is comprehensive, including maps, grids, line graphs, bar graphs, pie charts, surface grids, color grids, and spreadsheets.

You can create separate workbooks, for example, for different technologies and intersystem data. Each workbook can still contain several pages and frames, such as maps, layer 3 messages, decoded layer messages, idle mode data, connected mode data, etc. All workbooks and all pages within the workbooks are automatically synchronized in time. Nemo Analyze offers a wide range of ready-made workbooks with pages and data views for all relevant KPIs.

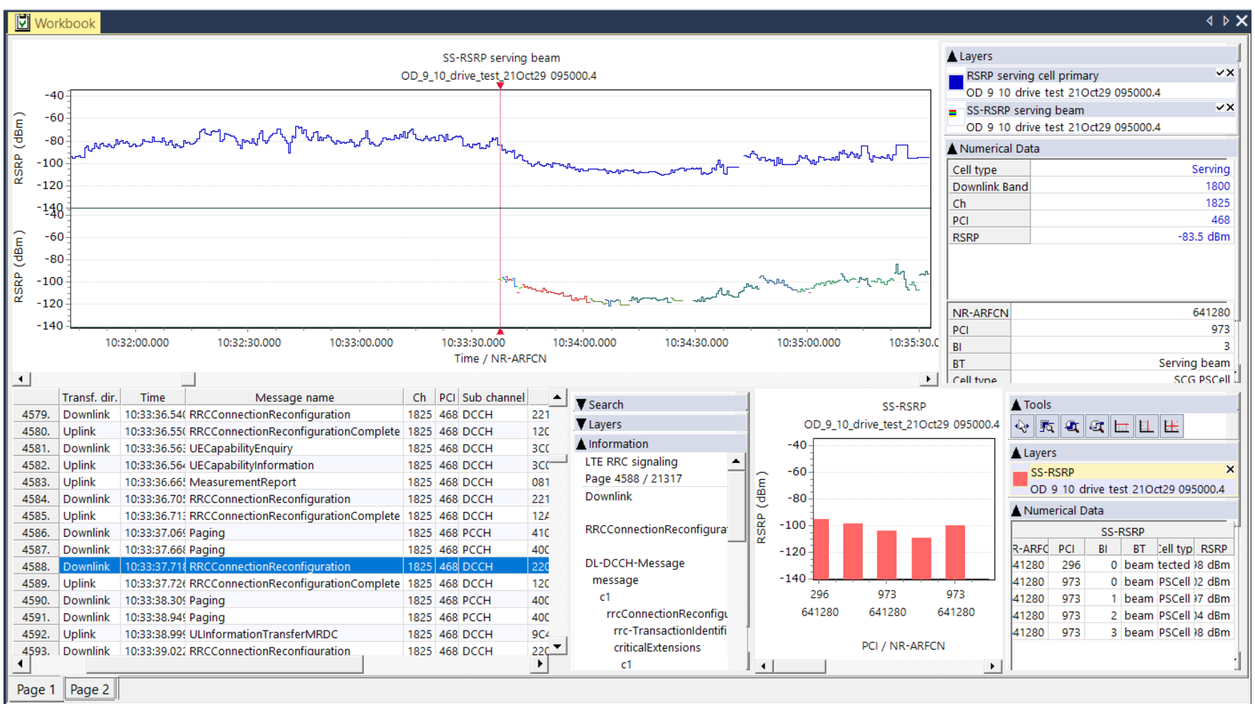


Figure 5. A multipage workbook with different data views – all of them time-synchronized.

Nemo Analyze graphs provide you with the full functionality generally expected of the featured graph types – and more. Color grids can be used with any two parameters, and with line graphs, you can have multiple layers, user-definable x- and y-axes, and data from different measurements and queries.

Nemo Analyze grids offer more detailed information on the measurement data. Grid data can be displayed in a numerical format in user-defined columns, and it can be saved as a delimited text file, exported to MapInfo format, or exported directly to MS Excel. You can also use color sets to highlight selected records in grids. Grids offer also extensive search capabilities, including searches for selected textual and numerical values from both grid records and decoded messages and several options for filtering and narrowing down the search results.

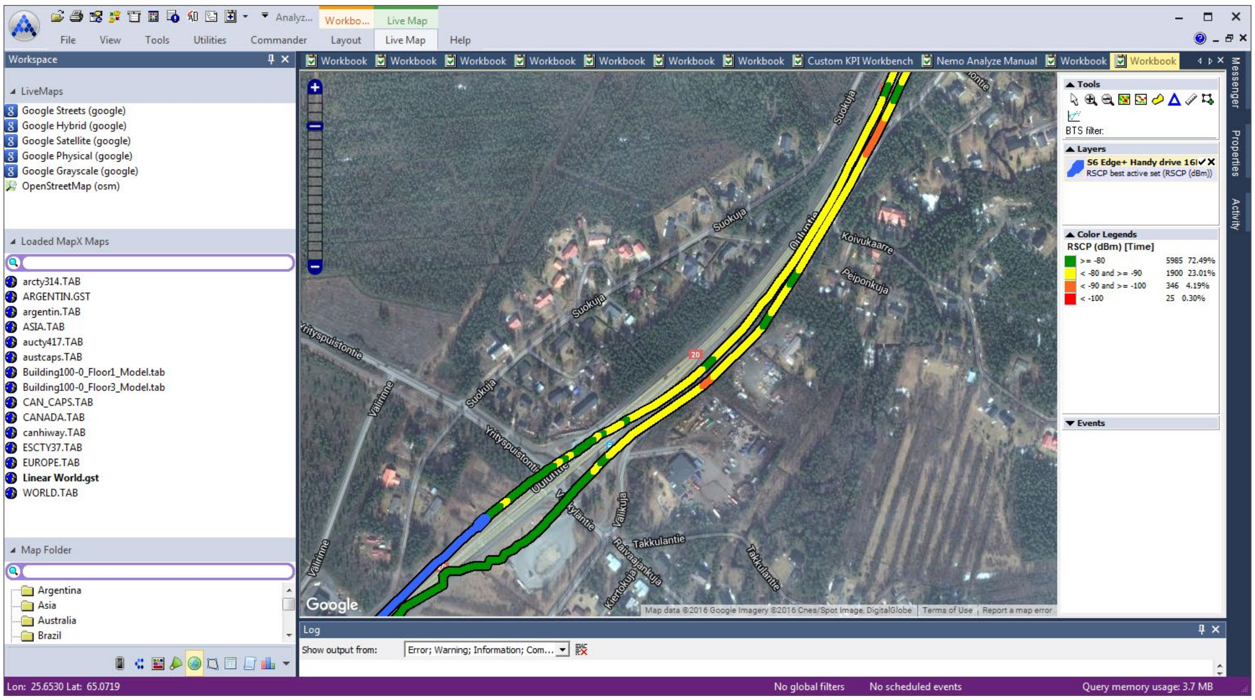


Figure 6. Live Google Map.

Nemo Analyze maps represent cutting-edge field test data visualization. Parameter-based route coloring enables you to find out at a glance the general picture of what your network performance is on a particular measurement route. The network parameter values along your measurement route are clearly visualized, which makes it easier, for instance, to single out problem areas for further analysis. It is also possible to compare two sets of measurements from the same route, with the color set reflecting the difference between the compared parameter values. You can use predefined color sets, or you can create your own, and you can set important events to be displayed on the map as icons.

Live maps (Google Maps and OpenStreetMaps) can be used the same way as regular MapInfo maps. Data and base stations can be plotted on the map and data can be dragged to the map as in MapInfo maps.

Nemo Analyze maps also provide you with an advanced base station overlay. With a user-defined BTS file, the map shows the location of each base station, the defined antennas with antenna directions, and even antenna apertures and cell identifying parameters. There can be multiple versions of the same base station file from different dates. Nemo Analyze selects the correct base station file version based on the queried measurement file automatically. The customizable base station reference files can also contain any user-defined information element not defined in the Nemo file format, such as RNC ID, SGSN ID, neighbor cells, etc.

Figure 7. Nemo Analyze reports and workbooks allow you to synchronize interactive base station maps with data views of any other type.

A line connecting the current location of the terminal or scanner to the serving cell (sector) will be drawn automatically (in UMTS systems, even multiple lines can be drawn to active sectors). This gives a highly visual impression of network operation, making it immediately evident if a call is connected to a non-optimal cell. Furthermore, when used together with problem-specific queries, the lines will point out the exact cells causing the problem (e.g., pilot pollution). Route coloring can also be used together with BTS files, the route coloring indicating the service areas and signal power of a selected antenna. With this feature, it is easy to optimize the service areas of single antennas. Base station map overlay can also be synchronized with grids containing base station data, i.e., the map will zoom automatically to the base station selected on the grid.

Area binning enables you to view statistics on a map. Each bin displays a parameter value aggregate (average, minimum, maximum, standard deviation, or count) as a color determined by the selected color set. Polygon area selection enables you to limit queries to a defined part of the route by simply selecting an area from the map. Area binning results can be exported as a MapInfo .tab file.

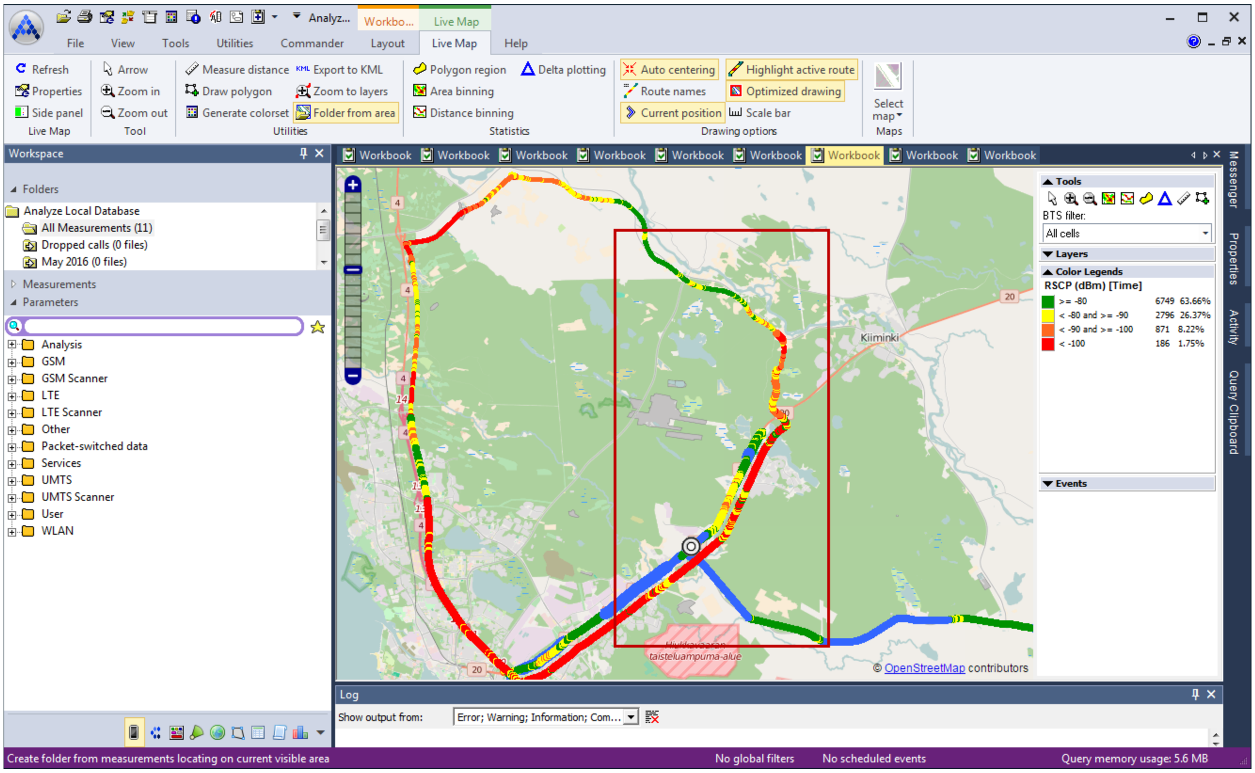


Figure 8. Area binning enables you to view vast amounts of measurement data on map.

Distance-based binning enables you to run statistics on a map based on route segments of your selection. In distance-based binning the bin is performed on the selected area by using distance values from the measurement files instead of coordinates. The workflow and bin result viewing are similar to area binning.

Nemo Analyze is MapInfo raster and vector map compatible. It also supports MapX Geoset files (.gst) that allow you to open several layers on a map and then save them all in a .gst file to be opened later. All user-defined map settings, such as the order of the different map layers and the zoom factor, are stored in the .gst file. It is possible to save maps as .jpeg images in user-defined map size.

Troubleshooting

Nemo Analyze offers a comprehensive set of troubleshooting KPIs and an advanced graphical user interface for custom KPI creation. Nemo Analyze also enables root cause analysis for voice calls, video calls, and for the UMTS RACH procedure, with failures and their reasons explicitly pinpointed. From the resulting data set that contains all network and application-level problems, you can select a problem and drill down to a user-defined time range of full event details surrounding the problem event. Because the database does not process any irrelevant data in a drill-down, problems can be found and analyzed quickly even from vast amounts of data.

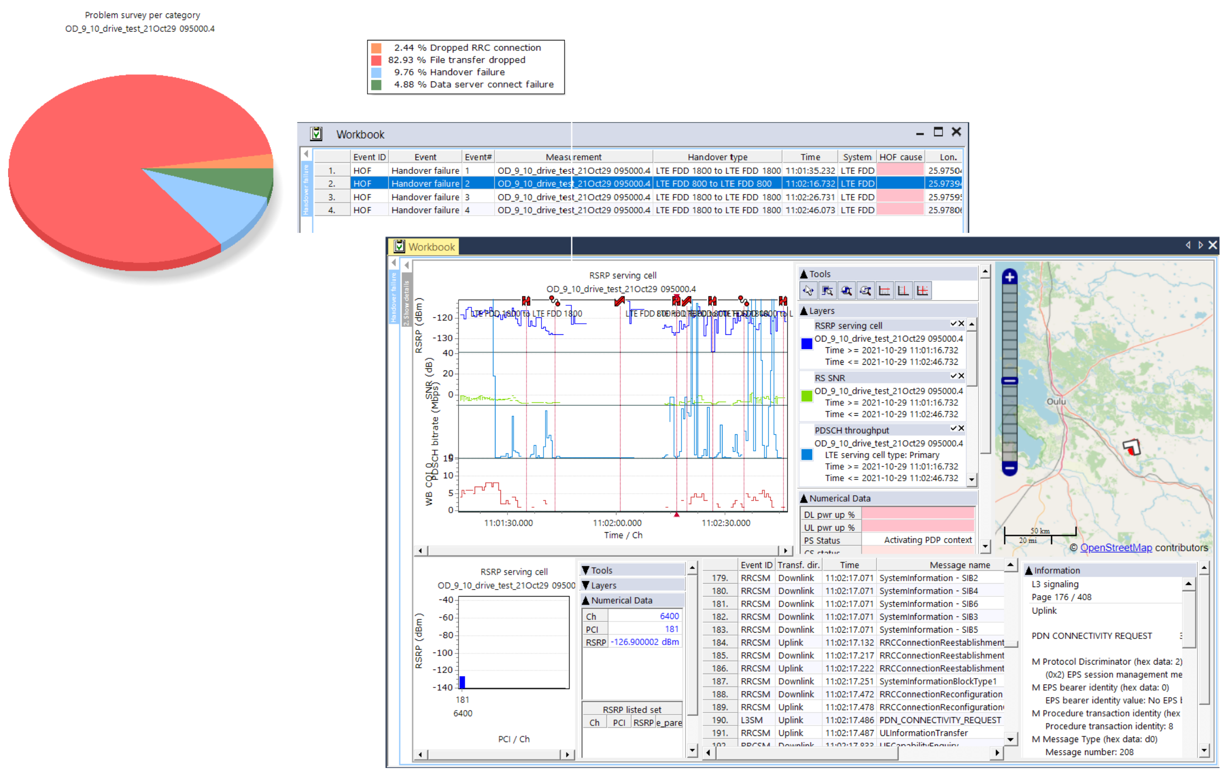


Figure 9. Drilling down from root-cause analysis to full event details.

The featured troubleshooting KPIs include LTE bad coverage and bad quality events, pilot pollution analysis (polluting and victim cells table with drill-down to map visualization), BCCH clash analysis (automatically finds co-BCCH interference cases from GSM scanner files), scrambling code clash analysis, call attempt failure analysis (root-cause analysis for GSM/UMTS call attempt failures), call drop analysis (root-cause analysis for GSM/UMTS dropped calls), RACH failure analysis (root-cause analysis for UMTS RACH failures), LTE/UMTS/GSM cross feeder/overspilling analysis, and UMTS 900 band clearance analysis (automatic detection and pinpointing of GSM interference sources in UMTS 900 band based on GSM scanner measurements). With the optional 5G Advanced Analytics feature, troubleshooting KPIs are complemented with NR bad coverage and bad quality events and NR cross feeder/overspilling analysis.

The KPI workbench makes it possible to create custom KPIs without knowing any SQL. It offers a graphical user interface for custom KPI creation, making it simply a matter of dragging and dropping the appropriate parameters, correlations, operations, and sort elements to the workbench view and connecting them into a logical flow chart. As also the properties of each of these elements are customizable, the combinations are almost limitless.

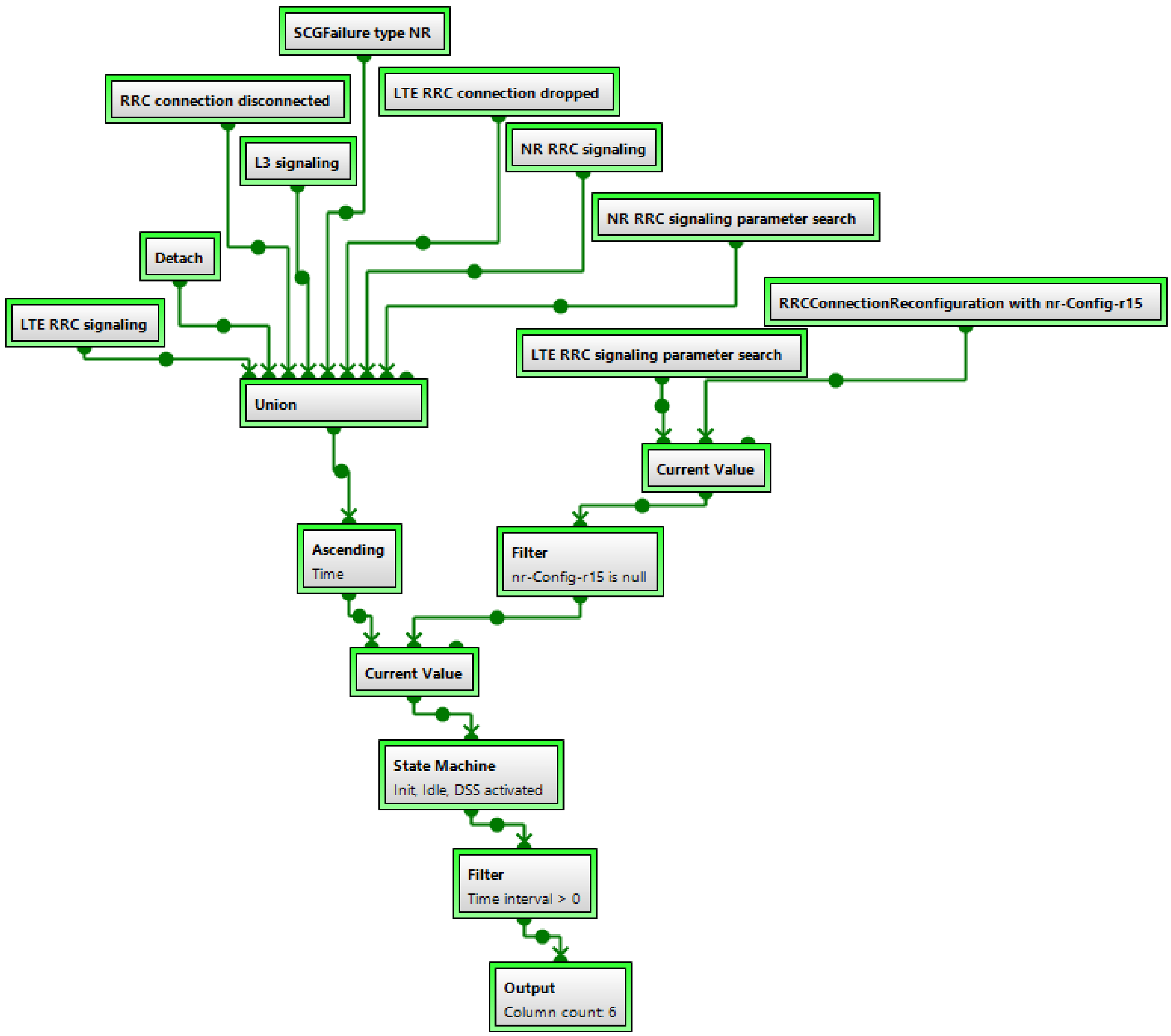
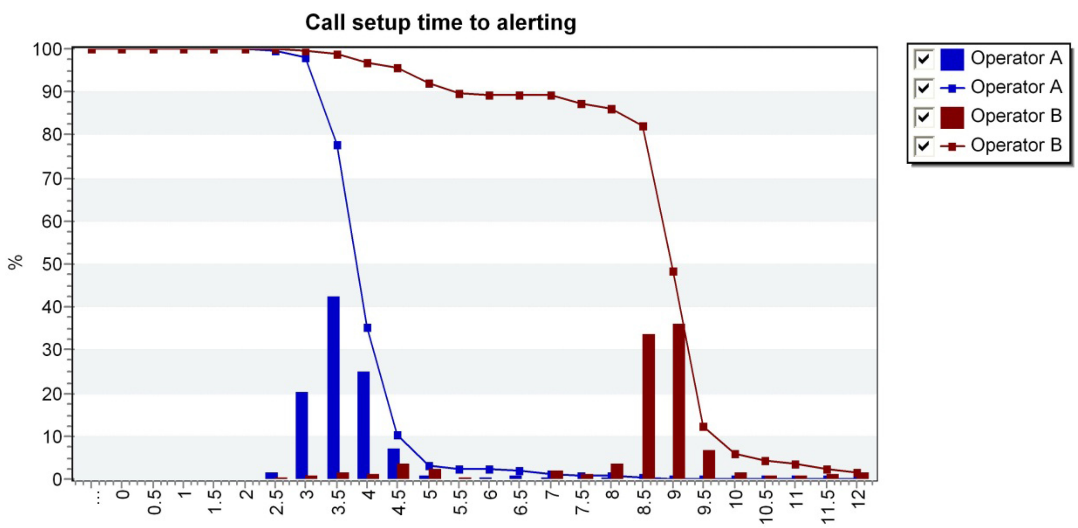


Figure 10. A complex custom KPI for dropped calls resulting from a missing handover.

Drill-down and KPI workbench are supported with all wireless technologies supported by Nemo Analyze. Root-cause analysis and troubleshooting KPIs are currently supported with 5G NR, LTE, UMTS, and GSM.

Benchmarking

Benchmarking with Nemo Analyze is easy and scalable to large data amounts, enhancing the cost and time efficiency of network operators. You can make data comparisons between different operators, technologies, cells, IMSIs, polygon areas and time frames, and view the results conveniently in a single graph or export data to complex reports. Benchmarking can be done over gigabytes of data for KPIs, such as voice call setup time, Ec/N0, Rx quality, and many more. Benchmarking can be performed quickly in the user interface for a single KPI. There are also predetermined benchmarking report templates that can be used in comparing KPIs from different operators, technologies, and time frames and in visualizing the results in a single report.



		Operator A	Operator B
Aggregates	Average	3.916	8.739
	Maximum	11.641	58
	Minimum	1.562	2.963
	Std. deviatior	3.916	8.739
	Variance	-15.333	-76.375
	Threshold <	96.703	8.006
	Sample count	546	637

Figure 11. Benchmarking of radio and application-level metrics over hundreds of hours of data.

Nemo Analyze provides the capability to report in CDR aggregated way (data is aggregated per transaction and further processed). Pre-defined report templates can be further customized.

With the optional NPS feature, you can report based on the Network Performance Score methodology where individual KPIs are aggregated and weighted to calculate an overall score.

Statistical Reporting

Nemo Analyze lets you select the optimal approach for each statistical reporting task. A set of predefined Microsoft Excel-compatible spreadsheet report templates for optimization and benchmarking purposes are available. The reports can be customized using the Nemo Analyze Spreadsheet Report Designer and MS Excel, and new custom reports can be easily created.

Nemo Analyze's Spreadsheet Report templates enable you to create custom report templates containing statistics, charts, and map plots of field test data in MS Excel format. Any metric from the parameter tree, including custom KPIs created with KPI Workbench can be used to produce source data for the report template. Nemo Analyze comes with default report templates for benchmarking and GSM, WCDMA, LTE, LTE-A CA, VoLTE, VoWiFi, IoT, and 5G NR optimization which can be modified with Nemo Analyze Spreadsheet Report Designer.

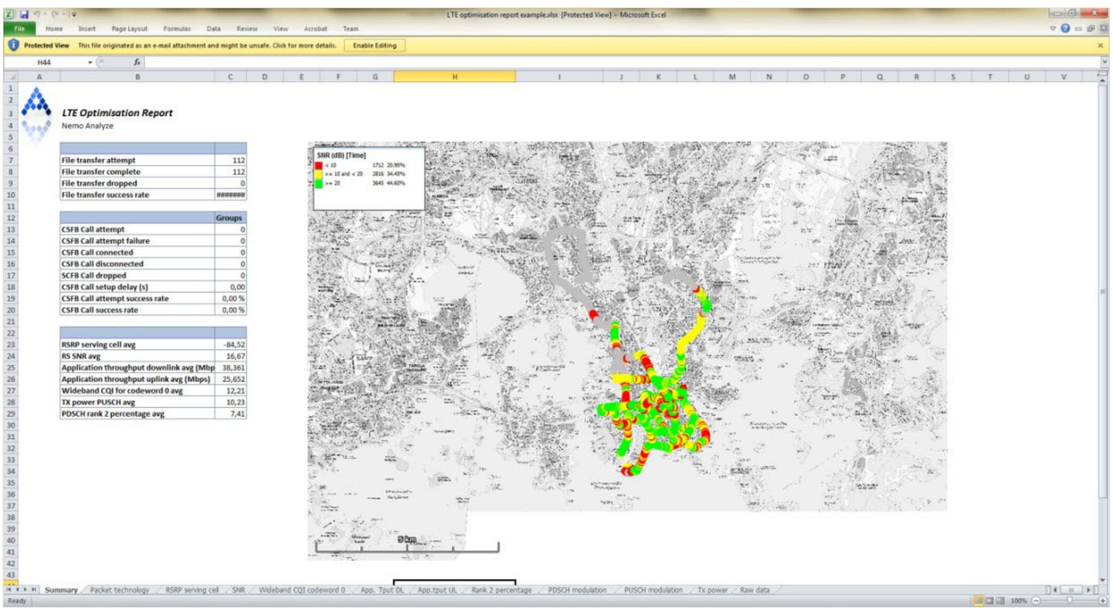


Figure 12. A summary page of a report created using Nemo Analyze Spreadsheet Report Designer.

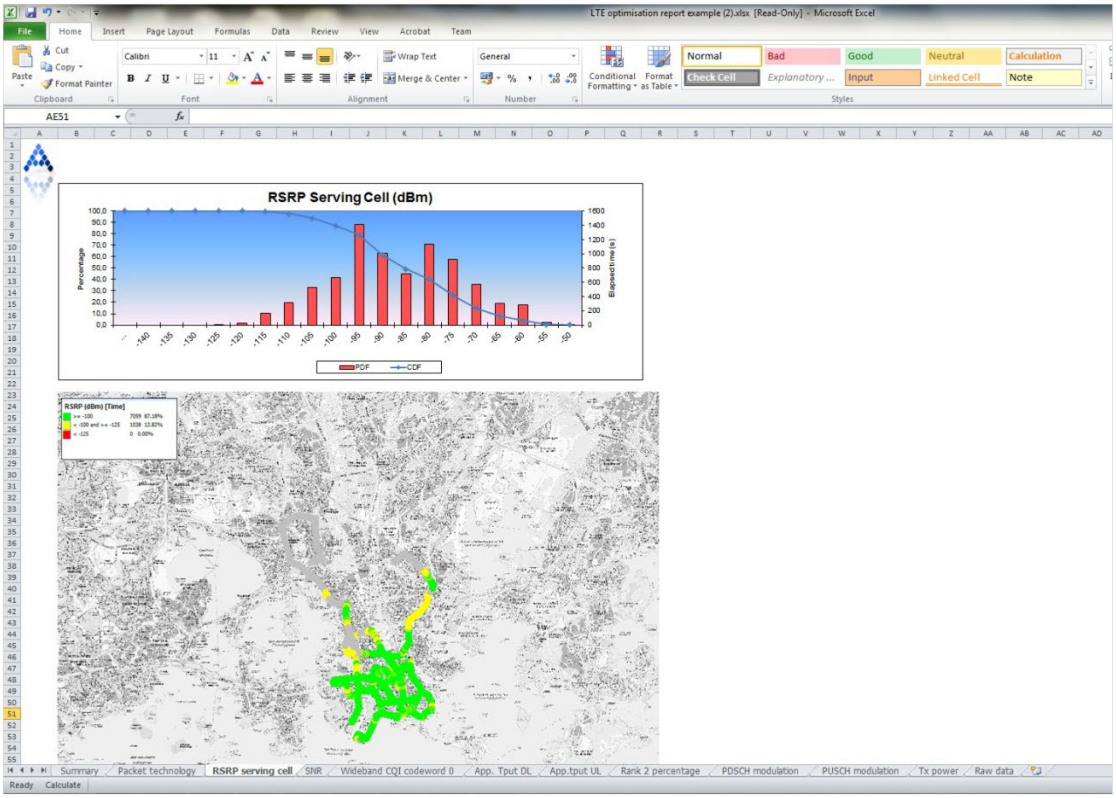


Figure 13. An example page of a report created with Nemo Analyze Spreadsheet Report Designer.

With Nemo Analyze's Spreadsheet Report Designer, you can create templates where you select parameters and statistics or create map plots in the desired design. The templates can be then run against any measurement data to generate the report for the selected data set. This feature allows you to also use the full data formatting functionality of MS Excel in reporting, and make use of any existing, third-party MS Excel spreadsheets you might already have.

Nemo Analyze introduces report templates in Microsoft PowerPoint (.pptx) and Microsoft Word (.docx) formats with limited features. Workbook page images of maps and charts of Nemo Analyze can be automatically exported to a .pptx or .docx document in predefined placeholders. Multiple images can be imported to the same pptx slide or docx document with user-defined size and aspect ratio. When there is a need to provide a standard report in .pptx or .docx format repeatedly on various DT campaigns, this functionality removes the manual work of copying and pasting pictures to PowerPoint or Word. You can create and save the workbook(s) for data views and maps to be used in the PowerPoint or Word report in Nemo Analyze, or you can use any existing Workbook template.